

MARKET BASKET ANALYSIS

Grocery Transactions · Apriori Algorithm · Association Rule Mining

Dataset: 9,835 transactions · 169 unique products · Python / mlxtend

Agenda

What we cover in this presentation

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Project Overview & Objectives

Dataset, tools, and goals of the analysis

02

Methodology

Apriori algorithm, encoding, rule mining

03

Dataset Statistics

Transaction count, product count, distributions

04

Top Selling Items

Support $\geq 2\%$: five dominant products

05

Frequent Itemsets

61 pairs at $\geq 2\%$; 3 pairs at $\geq 5\%$

06

Association Rules & Key Insights

Top 10 rules, lift analysis, business actions

Project Overview

Objective, dataset, and tools

OBJECTIVE

- ▶ Identify frequent product combinations in grocery transactions.
- ▶ Derive actionable association rules ($A \rightarrow B$) between product categories.
- ▶ Quantify rule strength using support, confidence, and lift metrics.
- ▶ Translate findings into shelf-placement and cross-sell recommendations.

DATASET

File: groceries.csv

Transactions: 9,835 baskets

Products: 169 unique SKUs

Format: ragged CSV (variable columns per row)

TOOLS & LIBRARIES

Python 3 · pandas · numpy

mlxtend (TransactionEncoder, apriori, association_rules)

matplotlib · seaborn

csv.reader (ragged array loading)

Methodology

From raw CSV to association rules

1

Load Data

csv.reader reads ragged CSV rows (variable columns) into a Python list of transactions.

2

Encode

TransactionEncoder converts transactions to a $9,835 \times 169$ binary matrix (1 = purchased).

3

Apriori Filter

Apriori algorithm prunes itemsets below 2% support, yielding frequent itemsets.

4

Generate Rules

association_rules() derives all $A \rightarrow B$ rules; each is scored on support, confidence, lift.

5

Rank & Interpret

Rules sorted by confidence; filtered by lift > 1 to identify positive associations.

Metrics: Support = $P(A \cap B)$ · Confidence = $P(B|A) = \text{Support}(A \cap B) / \text{Support}(A)$ · Lift = Confidence / Support(B) [Lift > 1 = positive correlation]

Dataset Statistics

High-level numbers from the grocery dataset

9,835

Total Transactions

169

Unique Products

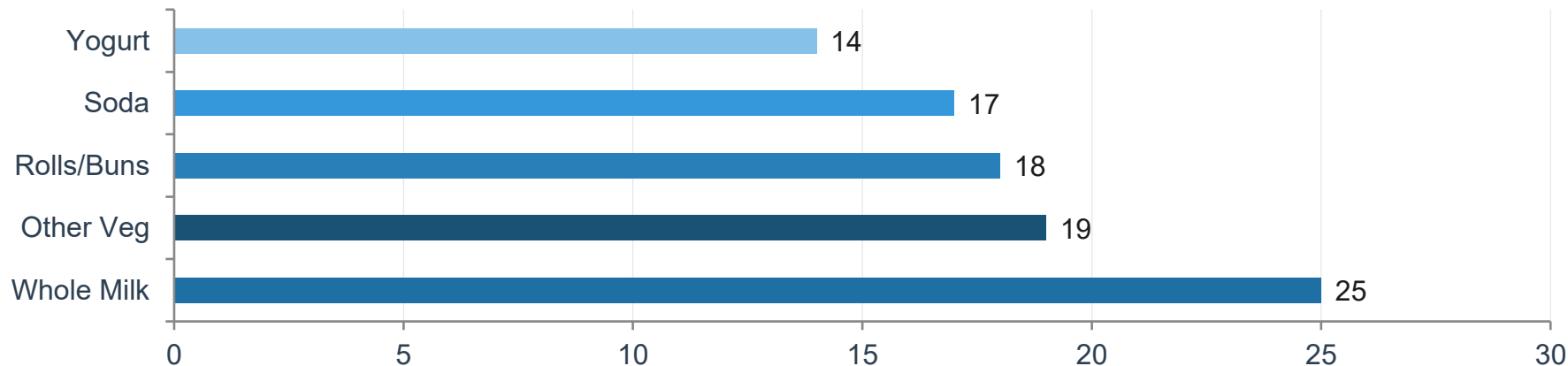
2%

Min. Support Threshold

126+

Rules with Lift > 1.0

Top 5 Products by Support (%)



Frequent Itemsets

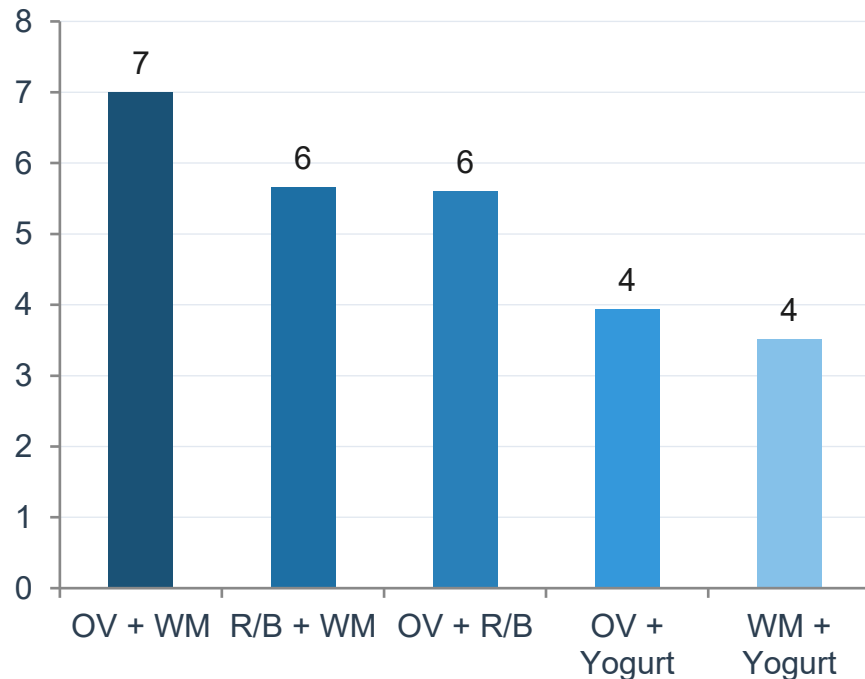
Product combinations exceeding minimum support thresholds

ITEMSETS WITH SUPPORT $\geq 5\%$

Itemset (length 2)	Support
Other Veg + Whole Milk	7.00%
Rolls/Buns + Whole Milk	5.66%
Other Veg + Rolls/Buns	5.60%

Only 3 itemsets clear the 5% threshold — joint purchase rate drops sharply beyond single items.

Top 5 Pairs by Support (of 61 pairs $\geq 2\%$)



OV = Other Veg · WM = Whole Milk · R/B = Rolls/Buns

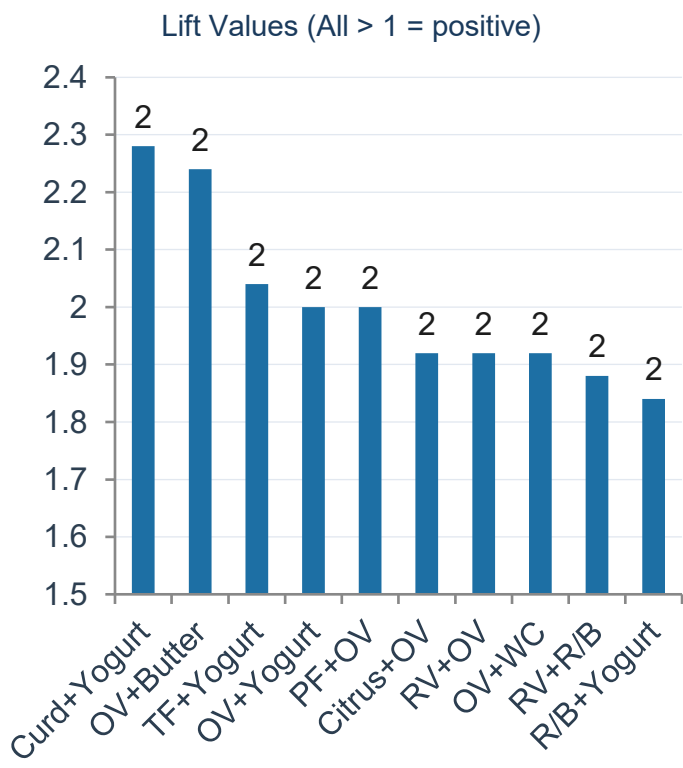
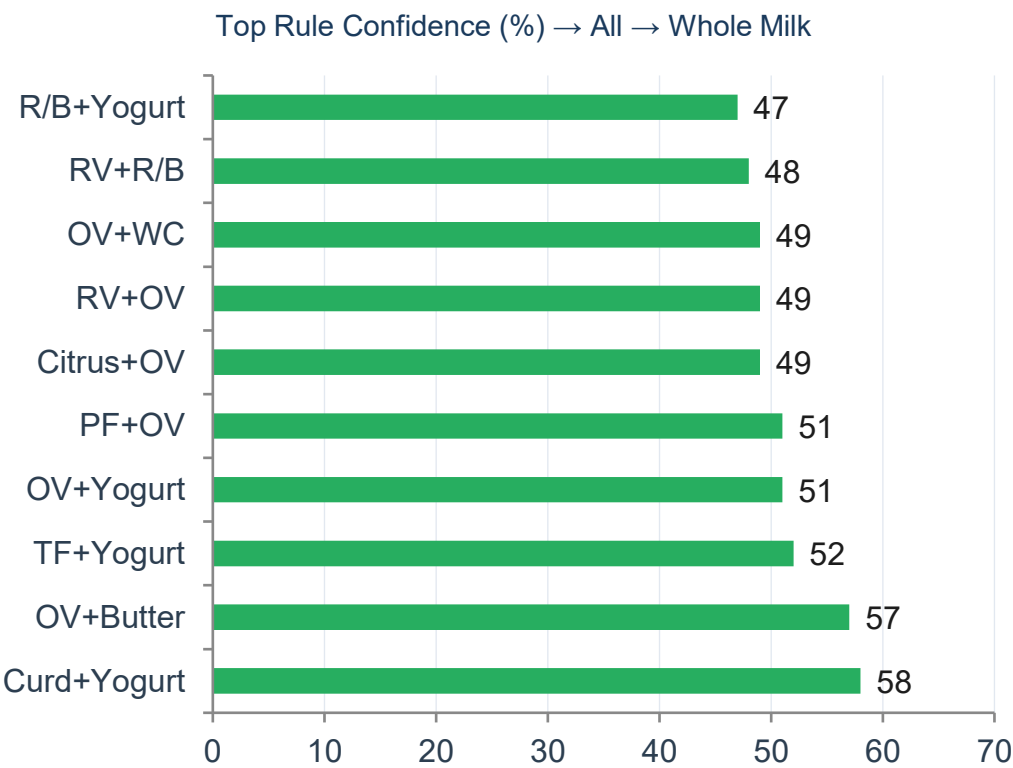
Top 10 Association Rules

Sorted by confidence (descending) · Min support $\geq 2\%$

Antecedent → Consequent	Rule	Support	Confidence	Lift
Curd + Yogurt	→ Whole Milk	2%	58%	2.28
Other Veg + Butter	→ Whole Milk	2%	57%	2.24
Tropical Fruit + Yogurt	→ Whole Milk	2%	52%	2.04
Other Veg + Yogurt	→ Whole Milk	2%	51%	2.00
Pip Fruit + Other Veg	→ Whole Milk	2%	51%	2.00
Citrus Fruit + Other Veg	→ Whole Milk	2%	49%	1.92
Root Veg + Other Veg	→ Whole Milk	2%	49%	1.92
Other Veg + Whip. Cream	→ Whole Milk	2%	49%	1.92
Root Veg + Rolls/Buns	→ Whole Milk	2%	48%	1.88
Rolls/Buns + Yogurt	→ Whole Milk	2%	47%	1.84

Lift & Confidence Analysis

126 rules with Lift > 1.0 and Support ≥ 2%



All 126 rules show Lift > 1.0, confirming positive correlations throughout the product catalog.

Business Insights & Recommendations

Actionable strategies from the analysis

Anchor Product Strategy

Whole milk (25% support) is the #1 consequent item. Ensure prominent placement and sufficient stock near high-affinity products like yogurt, curd, and other vegetables.

Cross-Sell: Veg–Dairy Bundle

Other Vegetables + Yogurt → Whole Milk has 51% confidence and lift 2.0. A bundled promotion or co-located display in the produce + dairy section can directly lift basket value.

Promotional Bundles

"Breakfast Bundle": Rolls/Buns + Whole Milk (5.66% joint support). "Dairy Trio": Curd + Yogurt + Whole Milk (58% confidence). Target these with volume discounts or combo pricing.

Store Layout Optimisation

Relocate whole milk end-caps near the produce aisle. Place yogurt and curd adjacencies within sight lines. Implement directional signage between these sections.

Key Takeaways

- 1 9,835 transactions:** 169 products analysed using the Apriori algorithm at 2% min. support.
- 2 Top product: Whole Milk:** Appears in 25% of all transactions; dominant consequent in association rules.
- 3 61 product pairs:** Cleared the 2% support threshold; 3 pairs clear 5%, led by Other Veg + Whole Milk (7%).
- 4 126 positive rules:** All rules with lift > 1.0 confirm catalog-wide cross-purchase dependencies.
- 5 Highest confidence:** 58% — Curd + Yogurt → Whole Milk — strongest actionable rule for promotions.